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(OCP) and over-temperature protection (OTP) are the typical protection functions of recent LDO products, as shown in Fig. 5. Besides protection functions as OCP or OTP, demand for additional functions has been rising recently in various application fields including consumer electronics.

First common demand for a linear regulator in recent consumer electronics is soft start function.

When an electronic system starts up, power supply voltage ramps up to a certain pre-defined voltage with a certain ramp rate (rise time). As power supply voltage rises, an inrush current flows into uncharged I/O capacitors.

The major problem of inrush current is that it exceeds absolute maximum current rating of applied components on the power rail. All components on the PCB have a specific current rating, which, if exceeded, could cause damage to these components. Likewise, all PCB traces are designed with a certain current-carrying capability in mind and are also at risk of damage.

When designing PCB traces and selecting components, not taking inrush current peak into account can damage the power path and lead to system failure; however, appropriately designing for a large inrush current peak will lead to increase in size and cost.

In viewpoint of linear voltage regulator, as presented in Fig. 5, inrush current can be reduced by increasing voltage rise time on load capacitance and slowing down the rate at which capacitors charge by applying internal soft start circuit and/or programmable soft start circuit.

Fig. 6 presents programmable soft start of the linear regulator by applying external soft start timing capacitor. With this feature, rise time can be programmed as per system requirement, as shown in Fig. 7, thereby reducing inrush current to ensure system stability.

PowerGood output is used as the power-on indicator to the load circuit. This signal indicates when output voltage is suitable for reliable operation of the load circuit. PowerGood output is pulled up or down with

respect to voltage level of output voltage.

Nowadays, PowerGood signal is used not only for output power-on indication but also for automatic power sequence. Fig. 8 presents an application example of PowerGood signal for power sequencing. When main powers (V_{IN1} and V_{IN2}) are applied, LDOs are ready to operate. After getting enable signal for the first LDO, the first LDO starts up and V_{OUT1} ramps up.

The internal PowerGood circuit monitors output voltage level of V_{OUT1} , and pulls up VPG1 signal when output voltage V_{OUT1} reaches the predefined voltage level. Then, this pulled-up VPG1 signal can be an enable signal of the second LDO, resulting in automatic power sequence between the first and second LDO. Power sequence delay time (enable delay time) is easily programmed by an external RC delay circuit.

As a result, it is quite easy to get the automatic power sequence timing diagram, as shown in Fig. 9. The complex power sequence as shown in the figure can be easily designed by simple wiring done using PowerGood signal, which provides design convenience for most consumer electronics.

Another demand of recent LDOs for consumer electronics is auto-discharge function. When LDO is turning off, internal auto-discharge function provides discharge path of output load capacitors, which are used for a stable power supply. As shown in Fig. 10, output voltage turns down much faster with auto-discharge function, helping in quick operation of consumer electronics, because an internal discharge path is connected internally through the LDO.

High-performance, high-reliability CMOS LDO products

To keep up with recent demands for linear regulators in consumer electronic equipment, overall performance of LDO regulators, such as power conversion performance, additional functions to adapt themselves into consumer electronics easily and reliability are being closely examined and proven. **EFY**